

Ankon Roy

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🌐 Portfolio 🔗 LinkedIn 🐙 GitHub

SUMMARY

Full-Stack Developer and final-year Computer Science student at KUET with a strong foundation in Data Structures & Algorithms. ICPC Dhaka Regionalist (2025) experienced in building scalable web applications from the database to the UI. Proficient in React, Next.js, Node.js, Spring Boot, and Laravel, with a proven ability to pick up new tools by building real-world projects across AI/ML, embedded systems, and blockchain.

EDUCATION

Khulna University of Engineering & Technology (KUET)

B.Sc. in Computer Science and Engineering (CSE)

Expected 2026

4th Year

EXPERIENCE

Frontend Developer Intern

Jul 2023 – Oct 2023

KuchToh Technologies Pvt. Ltd. (Remote)

- Completed a 2.5-month internship as a frontend developer within a fast-paced remote team.
- Built responsive frontend components and integrated REST APIs within a production codebase.
- Collaborated with senior developers to improve UI/UX and optimize web application performance.
- Participated in daily stand-ups, sprint planning, and code reviews using Agile workflows.
- Recognized as creative, hardworking, and a fast learner by the engineering team.

PROJECTS

Web Projects

B_Worm - Book Exchange Platform

Java 21, Spring Boot, PostgreSQL, Docker

- Community book exchange and donation platform with role-based access control.
- Implemented request/approval workflows, notifications, and an admin moderation area.
- Built using Java 21, Spring Boot, Spring Security, JPA, and Thymeleaf for server-side rendering.
- Containerized with Docker Compose for seamless local development and production deployment.
- Team collaboration using branch protection, code reviews, and a CI/CD pipeline.
- 🐙 GitHub | 🌐 Live Demo

Drink_Panda - Juice E-Commerce App

Next.js 15, Laravel 12, TypeScript, MySQL

- Full-stack e-commerce platform for ordering fresh juices online.
- Features include a dynamic cart, order tracking, and a comprehensive admin panel.
- Integrated bKash and Nagad payment gateways for seamless transactions.
- Frontend built with Next.js 15, React 19, TypeScript, and Tailwind CSS.
- Backend powered by Laravel 12 and MySQL, secured with Laravel Sanctum.
- 🐙 GitHub | 🌐 Live Demo

Library Management System

React, TypeScript, Laravel, SQL Server, Docker

- Comprehensive library system with distinct member and admin roles.
- Tracks borrowing records, fines, and inventory management.
- Includes an embedded Dialogflow chatbot for automated user assistance.
- Frontend built with React, TypeScript, and Vite; Backend with Laravel and SQL Server.
- Fully containerized using Docker and Nginx for consistent deployment.
- 🐙 GitHub

My Todo App

NestJS, MongoDB, EJS, Passport, Docker

- Task management application with per-user task isolation.
- Implemented secure session-based authentication using Passport.
- Backend built with NestJS (Node.js framework) and MongoDB.
- Containerized with Docker and deployed on Render.
-  GitHub |  Live Demo

Chat Application

Node.js, Express, Socket.io, MongoDB

- Real-time chat application built to master Express and WebSockets.
- Features instant messaging, user authentication, and persistent chat history.
- Utilized Socket.io for bidirectional real-time communication.
- Backend powered by Node.js, Express, and MongoDB with JWT authentication.
-  GitHub

EVOGYM - TypeScript React Fitness App

React, TypeScript, Tailwind CSS

- Gym landing page focused on high conversion and modern aesthetics.
- Built with React, TypeScript, and Tailwind CSS.
- Utilized Framer Motion for engaging scroll and hover animations.
- Implemented React Hook Form for user inquiries and membership sign-ups.
-  GitHub |  Live Demo

React Movie Info App

React, Redux Toolkit, SCSS, Axios

- Movie discovery app utilizing third-party APIs for real-time data.
- Features infinite scroll for seamless browsing and an embedded video player.
- State management handled globally using Redux Toolkit.
- Styled with SCSS and integrated Axios for API requests.
-  GitHub |  Live Demo

AI / ML / DL Projects

Battery Manager - AI Backend & ML Pipeline

FastAPI, PostgreSQL, XGBoost, LSTM

- AI-driven backend for battery telemetry ingestion and predictive maintenance.
- Built with FastAPI and PostgreSQL/TimescaleDB for efficient time-series data storage.
- Implemented multiple ML models: XGBoost (RUL), Isolation Forest (anomaly detection), K-Means (clustering), and LSTM (TTE prediction).
- Deployed on Render with CI/CD pipelines and REST APIs for predictions and advice.
-  GitHub |  Live Demo

Go Game with AI Agents

Python, Pygame, TensorFlow, NumPy

- Implementation of the classic board game Go with a Pygame interface.
- Features three distinct AI agents: Minimax with alpha-beta pruning, Monte Carlo Tree Search (MCTS), and a residual-CNN policy/value network.
- Developed a data-generation and training pipeline for the neural network model.
- Built using Python, Pygame, TensorFlow, and NumPy.
-  GitHub

Bangalore Home Price Prediction

scikit-learn, Flask, pandas, JavaScript

- First end-to-end machine learning project predicting real estate prices.
- Trained a regression model using scikit-learn and pandas on historical data.
- Served the model via a Flask REST API.
- Built a small, interactive web UI using JavaScript to consume the API.
-  GitHub |  Live Demo

Potato Disease Classification

TensorFlow, Keras

- Deep learning project for agricultural disease detection.
- Built a Convolutional Neural Network (CNN) to classify potato leaf diseases.
- Utilized TensorFlow and Keras for model architecture and training.
- Achieved high accuracy on the test dataset through image augmentation.
-  GitHub

Desktop Projects

Java Tutorial App

Java, JavaFX, FXML, Maven

- Desktop application for browsing programming courses.
- Features user login and signup functionality.
- Built using Java, JavaFX, and FXML for the UI.
- Managed dependencies and build lifecycle with Maven.
-  GitHub

Mobile App Projects

Note App Swift

SwiftUI, Firebase

- Native iOS notes application built with SwiftUI.
- Features user authentication and cloud-synced notes.
- Utilized Firebase Firestore for real-time data synchronization.
- Designed a clean, native iOS user interface.
-  GitHub

Other Projects

Cobra Language

C, Flex, Bison

- Designed and implemented a small programming language with custom syntax.
- Built a tree-walking interpreter to execute the language.
- Utilized Flex for lexical analysis and Bison for parsing.
- Implemented core compiler design concepts including AST generation and symbol tables.
-  GitHub

Fingerprint Attendance System

ESP32, AS608, MFRC522, Arduino, Supabase

- IoT-based attendance system using biometric and RFID verification.
- Utilized ESP32 microcontroller, AS608 fingerprint sensor, and MFRC522 RFID reader.
- Built a web dashboard for real-time attendance tracking.
- Integrated Supabase for cloud data storage and retrieval.
-  GitHub

ERC20 Token Project

Solidity, Hardhat, OpenZeppelin

- Developed a capped, burnable ERC-20 token on the Ethereum blockchain.
- Implemented a miner block reward mechanism for network incentivization.
- Utilized OpenZeppelin contracts for security best practices.
- Tested and deployed using Hardhat and ethers.js.
-  GitHub

Discord Server Management Bot

Python, discord.py

- Multipurpose Discord bot for community management.
- Features moderation tools, automated role and channel management, polls, and mini-games.
- Built using Python and the discord.py library.
-  GitHub

ACHIEVEMENTS

- **ICPC Dhaka Regionalist, 2025** – International Collegiate Programming Contest.
- Active competitive programmer on LeetCode, Codeforces, and CodeChef.

SKILLS

- **Data Structures & Algorithms:** Strong problem-solving foundation
- **Frontend:** React.js, Next.js, Redux, TypeScript, Tailwind, HTML5/CSS3, Framer Motion, SCSS
- **Backend:** Node.js (Express, NestJS), Python, Laravel (PHP), Spring Boot, Flask
- **Databases:** PostgreSQL, MySQL, MongoDB, SQL Server, Firebase, Supabase
- **AI/ML:** scikit-learn, pandas, NumPy, TensorFlow/Keras, spaCy, NLTK, XGBoost, LightGBM
- **DevOps & Tools:** Docker, Git & GitHub, Maven, Render, Vercel, Netlify
- **Others:** REST API integration, responsive design, Agile workflows, team collaboration